

UMF — Empirical Engagement

The observational programme for the Unified Metric Framework

Stephen L Browne | Central Coast NSW, May 2026 (AEST) | v1.3 | Supersedes v1.2 (18 May 2026)

Preamble

This document presents the empirical engagement programme for the Unified Metric Framework. It identifies the observational programmes through which the framework can in principle make contact with data, the specific signatures the framework predicts (where it can), and the empirical threads that remain at the structural level pending the resolution of open theoretical questions documented in the formal framework.

The framework is now publicly available at <https://github.com/OldGreyWhistle/The-UMF> as a citable reference; the current package is V2.3.

The empirical bar applied throughout is 5σ for proofs and 4σ for correlations, 2σ - 3σ notable. Findings reported below as "supportive" or "consistent with" are at the 4σ correlation level; findings at lower confidence are explicitly flagged as such, and at 2σ - 3σ the framework reports "not in tension with" rather than "alignment with."

The framework currently makes no quantitative predictions distinct from Λ CDM at the framework's rigour standard. At notable significance, the framework can identify directions of prediction consistent with observed trends (most importantly DESI DR2), but these are not claims of empirical support. The empirical engagement programme should be read as identifying the contexts in which the framework's future quantitative content would have to make contact with data, with one near-term decisive test (DESI DR3), one tractable first quantitative engagement (variable α through Bekenstein/BSBM formalism, recommended by both cold reviews), and one candidate first UMF-specific constraint-class analysis already undertaken without specialist input (bounded-universe topology from Planck CMB low- ℓ data, documented at the structural-inference level in the companion document UMF Bounded-Universe Topology v1.0).

Note on revision history: v1.1 (16 May 2026) introduced the "not in tension with DR2" framing and the dissipation→relaxation terminology change. v1.2 (17 May 2026) incorporated the reflective-boundary commitment's consequences (stochastic gravitational-wave background internal to Region A; bounded-universe topology constraint as candidate first analysis), and added reference to the framework lineage's prior empirical work under UMDT. v1.3 (18 May 2026) folds in the published bounded-universe topology positioning document, cites the specific ISW meta-analysis constraint in the framework lineage section, references the GitHub repository, and applies location and timezone consistency.

DESI DR2: not in tension with framework, decisive test pending DR3

DESI Data Release 2 (March 2025), reinforced by the DESI extended-analysis paper (DR2 + Planck + SNe, 2025) and the dynamical-DE Nature Astronomy paper (October 2025), detected a dynamical dark-energy signal at 2.5σ to 3.9σ depending on the analysis combination. The data favour a $w_0 w_a$ parameterisation with $w_0 \approx -0.84$ and $w_a \approx -0.7$, indicating $w(z)$ was below -1 (phantom) at moderate redshift and is now

approaching -1 from below, with the trend monotonic in time over the observed range.

The framework's position on DR2 is that it is not in tension with the framework's commitments. Several specific points apply:

- The DR2 signal is at notable significance (2σ - 3σ), below the framework's 4σ correlation bar. The framework therefore cannot claim alignment with DR2 under its own standing rule. The honest framing is "not in tension," meaning the framework's structural commitments are consistent with the direction of the DR2 trend, even though the trend is not yet at evidence-class significance.
- The direction of the DR2 trend ($w(z)$ approaching -1 from below over recent cosmic time, monotonic) is consistent with the framework's commitment to a relaxation process at the wavefront. The framework's mechanism is the universe relaxing toward primordial equilibrium; whether this produces $w(z)$ approaching -1 or departing from it depends on the specific relaxation mechanism, which is open theoretical work. UMDT-era empirical work (referenced in the framework lineage section below) suggests the framework's relaxation mechanism is at least consistent with the DR2 direction at notable significance, with one finding crossing 4σ correlation-class significance under UMDT's specific commitments.
- A 2σ notable finding pointing in the framework's direction is a reason to examine DR3 carefully when it arrives, not a reason to claim validation. The framework encourages further empirical examination at this level because it identifies a direction that DR3 may strengthen to correlation-class evidence.

DESI Data Release 3, expected in 2026-2027, is the decisive empirical test. The framework predicts:

- A signal reaching the 4σ correlation threshold or stronger.
- Monotonic functional form for $w(z)$: the equation-of-state parameter does not oscillate or reverse during the observed redshift range.
- Consistency with the DR2 direction ($w(z)$ approaching -1 from below, or some other direction the framework's relaxation mechanism predicts once specified).

The framework would be falsified, in its current form, by either:

- A confirmed oscillation or reversal in $w(z)$ at significant amplitude.
- A confirmed non-monotonic functional form that cannot be fit by any monotonic curve.

The framework would be in difficulty (not falsified, but requiring re-examination) by:

- A confirmed null result ($w(z) = -1$ at high confidence across the observed redshift range), which would force the framework to either commit firmly to plateau as the current epoch (with the DR2 signal being a statistical fluctuation) or to specify why the relaxation rate is below detection threshold.
- A reversal of the DR2 direction in DR3, which would be inconsistent with the relaxation commitment if the relaxation mechanism is monotonic in time.

Note that confirmed monotonic $w(z)$ deviation at 4σ would be consistent with the framework but would also be consistent with several other dynamical dark-energy models. The framework's commitment is that monotonic deviation in a specific direction (consistent with relaxation) is required; the framework does not currently have a specific functional form prediction that would distinguish it from competitors. This is one of the open theoretical questions in the Formal Framework document.

Bounded-universe topology constraint: first UMF-specific empirical engagement (filed)

The framework's commitment to a totally reflective inner boundary on the wavefront (Formal Framework v2.3) produces specific predictions for the cosmic microwave background at low multipole ℓ . If Region A is small enough relative to the surface of last scattering for CMB photons to make round-trip paths reflecting off the wavefront, the CMB temperature pattern should carry signatures of those round-trip paths — matched circles in the CMB sky, low- ℓ anomalies, or other topology-induced features.

A literature-review-and-positioning analysis at the structural-inference level has been undertaken without specialist input and is documented in the companion document UMF Bounded-Universe Topology v1.0. The analysis surveys the matched-circles methodology (Cornish, Spergel, Starkman 1998) and the existing Planck topology bounds (Planck 2013 XXVI, 2015 XVIII, 2018 VII; Vaudrevange et al. 2012; Bielewicz et al. 2011), surveys the Israel junction condition apparatus relevant to the wavefront's formal description (Israel 1966, Berezin-Kuzmin-Tkachev 1987, with contemporary peer-reviewed applications including Schubert 2026), and establishes the structural inference that UMF's bounded-universe and reflective-boundary commitments are not in tension with the existing Planck topology results, with the analysis at the structural-inference level pending specialist work to convert the inference into a quantitative constraint.

The five specialist work items identified by the bounded-universe topology analysis are: methodology adaptation (deriving the matched-circles signature analogue for reflective-boundary geometry), wavefront stress-energy specification (specifying S_{ij} under UMF's commitments using Israel junction conditions), quantitative constraint derivation (applying the adapted methodology to existing Planck data), Birkhoff theorem reconciliation (addressing the relationship between the reflective-boundary commitment and the Birkhoff prohibition on spherically symmetric gravitational radiation in standard GR), and pre-registered empirical engagement at correlation-class significance.

This is the first UMF-specific empirical engagement undertaken under the framework's standing rules without specialist input. The analysis was deliberately scoped at the structural-inference level rather than the quantitative-constraint level, after the literature review revealed that the methodology adaptation required for a quantitative analysis is properly specialist work. The deliverable strengthens the framework's position for the specialist engagement by establishing literature context and clear specialist targets, not by claiming quantitative results.

Variable α through variable ϵ_0 : the recommended first quantitative engagement (specialist)

Both cold reviews (ChatGPT and Claude Opus 4.7 incognito) independently identified the variable-coupling Maxwell sector as the most tractable first quantitative work item the framework can attempt with specialist input. The reasoning is that the existing literature (Bekenstein 1982, BSBM Sandvik–Barrow–Magueijo 2002, Olive–Pospelov 2002) provides established formalism for a scalar field coupled to the electromagnetic kinetic term varying ϵ_0 and hence α ; the existing observational bounds (atomic clocks, quasar spectroscopy, BBN) are tight; the work can proceed without committing the framework to a full gravity theory; and the framework either survives the comparison or is falsified in its variable- ϵ_0 commitment.

Outline of the engagement

The work would proceed approximately as follows, with each step subject to the framework's standing rule on full evaluation in UMF context before import.

- **Step 1.** Evaluate the Bekenstein/BSBM action as a candidate import. The action couples a scalar field ϕ to the electromagnetic kinetic term via $e^{(-2\phi)} F_{\mu\nu} F^{\mu\nu}$, with ϕ identified with (or as a monotonic function of) the framework's T_G . The evaluation must check units, regime of validity, and consistency with the framework's other commitments before adoption.
- **Step 2.** Verify that c-invariance is preserved. In BSBM the scalar field varies ϵ_0 and hence α , with c held constant by construction. The framework's c-invariance commitment is compatible.
- **Step 3.** Specify $\phi(z)$ (equivalently $T_G(z)$) over the relevant cosmological range using the framework's lifecycle phases. The trans-Planckian initial regime sets the boundary condition; the plateau and relaxation phases set the recent evolution.
- **Step 4.** Compute $\Delta\alpha/\alpha$ as a function of z . Compare to: (i) the ESPRESSO sub-ppm bound; (ii) the Murphy 2022 single-quasar bound; (iii) the Webb dipole result if taken at face value; (iv) atomic-clock present-day $d \ln \alpha / dt < 10^{-17}/\text{yr}$; (v) BBN constraints on α at $z \approx 10^9$ (typically $|\Delta\alpha/\alpha| < 10^{-2}$ at BBN, and tighter in some analyses).
- **Step 5.** Constrain the BSBM coupling strength. Report whether the framework's commitment to variable ϵ_0 is compatible with all five empirical bars, and if so over what region of parameter space.

If the answer comes back compatible across all five bars: the framework has a quantitative empirical engagement at the 4σ correlation bar, which it presently does not. If incompatible at 5σ : the framework's variable- ϵ_0 commitment is falsified in its current form, which is also a real result. Either outcome moves the framework off pre-paradigmatic status on this thread. This work requires the action-principle specialist input as a prerequisite (see Formal Framework Open theoretical questions).

Dark matter electromagnetic-propagation signature in SMBH environments

The framework's commitment to dark matter as gravitoelectric field contribution with a time-derivative component (in regions of time-varying gravitational driving) predicts an electromagnetic-propagation signature in environments where the gravitational dynamics are themselves time-varying. The natural observation targets are supermassive black hole environments, where accretion dynamics, jet launching, and time-varying orbital motion provide the time-varying gravitational driving.

The signature would manifest as a correlated electromagnetic anomaly in SMBH environments that would not be present in static dark matter configurations but should be present where dark matter is responding to time-varying gravitational driving. The specific observable form of the signature is open and requires specialist input. Candidate signatures include:

- Correlated drift in atomic transition frequencies in absorbing or emitting material near SMBHs, beyond the standard gravitational redshift contributions.
- Anomalous impedance variation in radio-band observations of SMBH jets and accretion flows, beyond the standard plasma physics.
- Time-correlated variability in spectroscopic features across spatially separated regions of SMBH environments, indicating coordinated variation driven by the time-varying dark matter response rather than by local astrophysical processes.

These candidate signatures are at the structural-prediction level and have not been worked out at the level of quantitative prediction. The framework's position is that the signature should exist; the specific form requires specialist input on the static-to-dynamic dark matter transition and on the variable-coupling Maxwell equations.

Stochastic gravitational-wave background under the reflective-boundary commitment

The framework's commitment to a totally reflective inner boundary on the wavefront has consequences for the primordial gravitational signature observable inside Region A. Primordial gravitational waves from the establishing event do not propagate freely out to infinity; they reflect off the wavefront's inner surface and remain bounded inside Region A. Because the primordial pulse is stochastic in character (a single chaotic event rather than a coherent periodic source), the reflected gravitational content forms a stochastic gravitational-wave background internal to Region A rather than a coherent standing-wave structure.

This framing replaces the earlier framework versions' "monopolar gravitational-wave echo" formulation. The stochastic background framing partially defuses the Birkhoff theorem issue Claude Opus 4.7 cold review flagged: a stochastic gravitational-wave background is not spherically symmetric in the relevant sense (individual modes have arbitrary angular structure), so the Birkhoff prohibition on spherically symmetric gravitational radiation does not apply directly.

The framework's prediction is a stochastic gravitational-wave background with characteristic frequency content set by the primordial pulse's spectrum and modified by the wavefront's reflective boundary. The signature's observability depends on its frequency range and amplitude. The natural frequency scale for primordial echoes inside a Hubble-scale Region A is set by the round-trip path across Region A, which gives frequencies many orders of magnitude below the nHz band of pulsar timing arrays. NANOGrav data is therefore ruled out as a test of the framework's SGWB signature, on two grounds: the Nyquist constraint (PTAs cannot resolve frequencies below their inverse observation baseline) and the BIPM clocking concerns (PTA timing model corrections may absorb signatures of the kind the framework predicts).

The framework does not currently specify the spectrum of the SGWB or its amplitude. This requires specialist input on the establishing pulse's character and on the reflective-boundary modification of the spectrum. The SS-TLGO design concept (April 2026) developed in the framework's supporting documents specifies an instrument capable of measuring absolute phase rather than differences, which would in principle be sensitive to stochastic signatures at frequencies below the differential-detection regime. The SS-TLGO concept is at the 50-100 year horizon and is not a near-term observable.

Indirect manifestation through correlated drift in fundamental constants

The stochastic SGWB inside Region A may manifest indirectly through correlated drift in fundamental constants. The mechanism would be that SGWB modes modulate local T_G uniformly across the spatial region they propagate through, which produces correlated variation in ϵ_0 (and consequently α) through the framework's variable-coupling commitments. The signature would appear as correlated drift across many independent observable systems: atomic clock comparisons, nuclear decay rate measurements, and quasar spectroscopy at different redshifts.

This is the empirical thread that connects the SGWB to currently accessible data. The signature is the correlated drift across independent systems that would not be expected if each system's variability were driven by local astrophysical processes. The analysis would require pulling together data from heavy-isotope half-life measurements, atomic clock comparison archives, and quasar spectroscopy across multiple instruments and time periods, looking for correlated drift signatures.

The analysis is computationally tractable but requires specification of the functional form of $\epsilon_0(T_G)$, of the expected magnitude of SGWB modulation, and of the spectrum of the SGWB. All three are open theoretical questions. The empirical thread is therefore at the candidate level — the analysis can be designed but the quantitative predictions to test against cannot be made until the theoretical work is done. The framework lineage's methodology limits noted for PTA observations (Nyquist constraint, BIPM clocking concerns) may apply analogously to atomic clock and quasar spectroscopy data; the methodology question is best evaluated by the specialist before the analysis is attempted.

CMB acoustic peaks: an issue the framework must address

The CMB acoustic peak structure (Planck 2018, ACT, SPT) requires non-baryonic dark matter to be gravitationally clustering by recombination at $z \approx 1100$, before stars or galaxies exist. The framework's gravitoelectric DM is described as "sourced by the wavefront and modulated locally by the gravitational distribution of baryonic matter (galactic concentrations, including SMBHs where present)." Galaxies and SMBHs do not exist at $z = 1100$.

The Claude Opus 4.7 cold review correctly identified this as the standard hard test that modified-gravity / no-particle DM programmes (MOND, TeVeS, MOG, Verlinde's emergent gravity) face. The framework's structural account inherits the challenge. The framework needs to specify how the gravitoelectric DM contribution behaves at recombination, when there are no galactic gravitational concentrations to provide local modulation.

The framework's candidate response is that the wavefront sourcing of the gravitoelectric DM contribution operates from the establishing event onward, independently of whether baryonic structures are present. The local modulation by baryonic concentrations is a refinement that becomes relevant after structure formation; before structure formation, the wavefront sourcing alone determines the DM contribution. This is a structural commitment that has not been worked out quantitatively; it does not yet address whether the resulting DM contribution at recombination produces the observed acoustic peak structure.

The framework's open theoretical questions list now includes a related item on potential simultaneous resolution of the S_8 clumpiness deficit, BBN light-element abundances, and CMB acoustic peaks tensions through DM behaviour. The three observationally distinct tensions may share a common dependence on UMF's gravitoelectric DM behaviour at recombination and later epochs; the specialist input is the natural context for evaluating whether UMF's commitments produce a coherent simultaneous resolution.

The author has indicated existing UMDT-era empirical work, including a CMB analysis, that may be relevant to this question. This is to be incorporated in a future revision once summarised. Pending that incorporation, the framework's position is that the CMB acoustic peaks are a serious open empirical question that the framework must address, with the specialist engagement being the natural context in which the address is worked out.

Acknowledgement: the Bullet Cluster

The Bullet Cluster (1E 0657-558) is widely cited as the single strongest empirical case for dark matter being a distinct entity from baryonic matter. The standard

interpretation places dark matter as a collisionless species that passed through the galaxy cluster collision while the gas was slowed by ram pressure, producing the observed offset between the lensing peaks (tracking total mass) and the gas (tracking baryonic mass).

The framework's account of the Bullet Cluster is structural rather than quantitative. The single-species gravitoelectric dark matter, characterised by its contribution to the gravitoelectric field with the field shaped by the gravitational distribution of baryonic concentrations, naturally produces the observed phenomenology qualitatively: the galaxies pass through the collision largely undisturbed (because galaxies are mostly empty space), the dark matter field is shaped by the galactic gravitational distributions and stays with them, and the lensing signal peaks at the galaxy locations rather than at the gas location.

The Claude Opus 4.7 cold review correctly flagged that this structural account is inherited from MOND/TeVeS class theories, which have had quantitative difficulty matching the observed lensing offset (40-80 kpc). The framework's account inherits both the structural picture and the inherited quantitative difficulty. The framework cannot calculate the magnitude of the lensing offset, the angular separation between the lensing peaks and the gas, or the specific shape of the lensing signal without specifying the formal apparatus for the gravitoelectric dark matter contribution. The Bullet Cluster is acknowledged as a major cosmological observation that the framework's account addresses structurally but does not yet engage quantitatively, and the inherited difficulty is noted.

Framework lineage: prior empirical work under UMDT

The framework lineage has conducted five pre-registered empirical analyses under the UMDT precursor framework. The analyses are referenced here as context for the framework's empirical methodology, not as direct support for UMF's specific commitments. UMDT and UMF differ in their structural commitments (UMF's T_G as independent dynamical field with reflective boundary; UMDT's parametrised impedance components), so the UMDT analyses inform UMF's empirical capacity without validating UMF directly.

The five UMDT analyses are:

- **Pantheon+ residual spatial analysis (April 2026).** Pre-registered analysis of 1580 Type Ia supernovae for localised G-tension variation signatures. All five pre-registered tests returned null at the 3σ Bonferroni-corrected threshold. Converted to an upper limit on coherent fractional metric variation of approximately 1.1% in the frequency band above 5.8×10^{-18} Hz. First numerical constraint on a UMDT-specific prediction.
- **ISW cross-correlation meta-analysis (April 2026).** Meta-analysis of six published ISW measurements spanning 2008-2022 (Ho et al. combined 2008, Giannantonio et al. combined 2008, Goto et al. WISE 2012, Giannantonio et al. updated 2012, Krolewski & Ferraro unWISE 2021, Bahr-Kalus et al. RACS 2022). Weighted mean $A_{\text{ISW}} = 1.04 \pm 0.13$, consistent with Λ CDM at the 0.3σ level. Marginal 1.49σ linear redshift trend was driven by a single Goto 2012 outlier subsequently superseded by improved WISE data analyses. Converted to a constraint of $|q| < 0.3$ at 3σ on any linear redshift trend of the form $A_{\text{ISW}}(z) = 1 + \alpha \cdot z$. Any future UMDT or UMF formulation producing a specific numerical prediction for dynamical dark energy amplitude evolution in the ISW channel will be testable against this constraint.
- **CMB echo parametric pumping.** Null result.
- **CMB lattice Path 1.** Null result.

- **DESI DR2 BAO analysis (April 2026).** Pre-registered analysis with four tests; all four pass at pre-registered significance thresholds. The P2 phantom-crossing preference reaches 4.2σ correlation-class significance for the DESI+CMB+DESY5 combination. First positive consistency match in the programme.

The methodology across the programme is consistent: pre-registration before data inspection, Bonferroni correction for multiple testing, null reporting as primary deliverable, honest caveats. The four nulls plus one positive consistency match (with one finding crossing 4σ correlation-class significance under UMDT) is the empirical record of the framework lineage.

This empirical record is context for the specialist engagement. It demonstrates that the framework lineage has capacity for pre-registered empirical analysis at the methodological standard the field expects, and that UMF-specific pre-registered analyses can in principle be undertaken once the framework's commitments (action principle for T_G , $\epsilon_0(T_G)$ functional form, gravitoelectric DM formalism) are tight enough to permit specific predictions. The bounded-universe topology positioning document (v1.0) is the first UMF-specific empirical engagement at the structural-inference level, undertaken without specialist input under the standing rules.

What the framework does not currently engage

For honesty and to set appropriate expectations, this document lists empirical programmes that the framework does not currently engage. These are not necessarily outside the framework's scope; they are observations for which the framework does not currently have a specific prediction or which it cannot currently address at the rigour standard.

- **The H_0 tension.** V1.5 attempted to address this through spatial coupling variation; the attempt produced a parameter-regime bound violation against BBN and LLR constraints. The framework does not currently offer an alternative mechanism. The H_0 tension remains an open question.
- **CMB power spectrum quantitative details.** Beyond the acoustic peaks issue flagged above, the framework does not currently make predictions about the specific features of the CMB power spectrum that distinguish it from Λ CDM. UMDT-era CMB analysis to be incorporated in future revision.
- **Inflation and the horizon problem.** The framework's establishing event is gestural and does not provide a mechanism for the observed CMB homogeneity across causally disconnected patches, the observed near-flatness, or the observed scale-invariant primordial power spectrum. The framework must either adopt inflation (clarifying how it sits inside the Region A / B / C structure) or propose an alternative mechanism. Flagged for specialist input.
- **Large-scale structure formation.** The framework does not currently make specific predictions about how large-scale structure forms differently from Λ CDM. The single-species gravitoelectric dark matter is structurally consistent with the observed large-scale structure but the quantitative predictions are open pending specialist input on the gravitoelectric formalism.
- **Galactic rotation curves at quantitative level.** The framework reproduces the qualitative phenomenology of flat rotation curves through the gravitoelectric dark matter contribution, but it does not yet produce the specific functional form (NFW profile, Einasto profile, or alternative) without specification of the formal apparatus. The baryonic Tully-Fisher relation and other galactic-scale tight correlations are similarly at the structural level only.
- **JWST early massive galaxies.** Recent JWST observations of massive evolved galaxies at $z > 10$ challenge standard Λ CDM structure formation timelines. UMF could in principle engage this — the framework's commitments may produce different initial

conditions for structure formation — but the engagement is properly specialist work because the inferred stellar masses depend on assumptions UMF would have to evaluate before adopting. Flagged for specialist input.

- **S_8 clumpiness deficit (weak lensing tension).** The $2-3\sigma$ tension between weak lensing surveys (KiDS, DES, HSC) and CMB-derived predictions of late-time matter clustering is at notable significance only. UMF's gravitoelectric DM commitment could in principle produce different clustering behaviour, but specifying this requires the formal apparatus for gravitoelectric DM that is open theoretical work. This tension is referenced in the Formal Framework open question on potential simultaneous resolution with BBN and CMB acoustic peaks.

This list is not exhaustive. The framework does not currently engage observation at the level of specific predictions in most of the active cosmological observational programmes. This is consistent with the framework's pre-paradigmatic status and is not in itself a problem; it is a description of where the framework currently is in its development cycle.

Summary

The framework's empirical engagement programme is centred on one near-term decisive test (DESI DR3 monotonic $w(z)$ at correlation-class significance), one filed UMF-specific constraint-class engagement at the structural-inference level (bounded-universe topology from Planck CMB low- ℓ data, documented in the companion document UMF Bounded-Universe Topology v1.0), one tractable first quantitative engagement requiring specialist input (variable α through Bekenstein/BSBM formalism, recommended by both cold reviews), one empirical thread (dark matter electromagnetic-propagation signature in SMBH environments), one structurally-but-not-quantitatively engaged observation (the Bullet Cluster), one structural prediction requiring further theoretical development (the stochastic SGWB under the reflective-boundary commitment), and one issue the framework must address with the help of existing UMDT-era empirical work (CMB acoustic peaks at recombination).

The honest framing is that the framework is at the boundary of being able to engage observation at the rigour standard. The framework is not in tension with DR2 at notable significance, and DR3 will be the first opportunity for the framework to engage observation at correlation-class significance. The bounded-universe topology positioning document is the first UMF-specific empirical engagement under the standing rules, at the structural-inference level. The variable- α work using Bekenstein/BSBM formalism is the cleanest first quantitative engagement with specialist input, and both cold reviews recommend it as the priority work item.

The framework is not currently competitive with Λ CDM at the level of specific quantitative predictions across the active observational programmes. The framework's strength is structural: it offers a coherent alternative picture of the cosmological lifecycle that, if developed to the level of formal apparatus, could engage observation at the precision level. The development to that level is the work the specialist engagement is intended to advance, with the bounded-universe topology positioning document at the structural-inference level as the near-term engagement that has been undertaken without specialist input.